THE COMPLETED TENSOR PRODUCT OF WEAKLY PRE-ADIC RINGS IS WEAKLY ADIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to the proof of the preceding lemma rather than to its statement alone; that reference is carried as a navigation stub and recorded as a prerequisite.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The characterisation of weakly adic and weakly pre-adic rings. Same chapter, Lemma lemma-weakly-adic, invoked at proof lines 1762-1764. Reduces the assertion to admissibility plus openness of the closures of the powers of an ideal of definition.
- Admissibility of the completed tensor product, and the ideal of definition obtained in its proof. Same chapter, Lemma lemma-completed-tensor-product, invoked at proof lines 1764-1770. Supplies admissibility and the ideal of definition whose power closures have to be shown open.

Source reference. Original lemma 0GB4: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1187–1187. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1633–1633. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $B \to A$ and $B \to C$ be continuous homomorphisms of linearly topologized rings. If A and C are weakly pre-adic, then $A\widehat{\otimes}_B C$ is weakly adic.*

Proof. We will use the characterization of Lemma 0GXF without further mention. By Lemma 0GB4 we know that $A\widehat{\otimes}_B C$ is admissible. Moreover, the proof of that lemma shows that the closure $K \subset A\widehat{\otimes}_B C$ is an ideal of definition, when $I \subset A$ and $J \subset C$ of $I(A\widehat{\otimes}_B C) + J(A\widehat{\otimes}_B C)$ are ideals of definition. Then it suffices to show that the closure of K^n is open for all $n \geq 1$. Since the ideal K^n contains

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

$I^n(A\widehat{\otimes}_B C) + J^n(A\widehat{\otimes}_B C)$, since the closure of I^n in A is open, and since the closure of J^n in C is open, we see that the closure of K^n is open in $A\widehat{\otimes}_B C$. $\square$